

Summary Report



Department of Physical Sciences

**An Analytical and Computational Approach
to the Scotogenic Model of Neutrino Mass Generation and
Dark Matter**

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Abstract

In this project, we shall be working towards our aim to understand Scotogenic model of Dark Matter and Neutrino Mass generation as well as trying to apply that model and getting results. For this we shall be referring mainly to paper by Ernest Ma[1], then formulating the model using S.A.R.A.H.[2], a Mathematica package and SPheno[3], [4] then implementing on MicroOmegas which is applicable mainly on SupperSymmetric Dark Matter[5], finally implementing on Collider Data using MadGraph[6]. We shall be taking deviations as required.

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Chapter 1

Deviation I-Neutrino Mass Mechanisms[7], [8]

Since standard model predicts neutrino masses to be zero which has been seen not to be true in many experiments and astroparticle phenomena, we need to generate neutrino masses by some way or other.

The simplest way to generate neutrino masses is by using 5-dimensional Weinberg operator but a little is known about it, moreover it violates lepton number. One of the most famous mechanism is SeeSaw Mechanism which we will be focusing on this section, since the article we referred to[7] discusses it most. In a future chapter we shall be focusing upon Scotogenic Model.

Since we need to extend our Standard Model to account for Neutrino masses, we can take one of the 3 big routes;

- SeeSaw Mechanism
- Extra Dimension
- R-parity violating SUSY

Here, we shall be focusing on SeeSaw assuming that there exist 3 active neutrinos to account for 2 large mixing angles and 1 small mixing angle as well as pattern of neutrino masses consistent with experiments. We shall be strongly emphasizing on models where 2 large mixing angles can arise naturally and consistently with neutrino mass heirarchy.

Sequential Dominance: If one of the the right-handed neutrinos contributes dominantly in the see-saw mechanism to the heaviest neutrino mass and a second right-handed neutrino contributes dominantly to the second heaviest neutrino mass, then large atmospheric and solar mixing angles may be interpreted as simple ratios of Yukawa couplings. if ν_R contributes equally then there is no Sequential Dominance.

1.1 Contribution to neutrino masses and mixing angles by charged leptons

It can be noticed that in general neutrino masses and mixing angles can originate atleast in part from charged lepton sector. writing low Energy Lagrangian involving charged lepton and neutrinos mass matrices

$$\mathcal{L}_{\text{mass}} = -(\bar{e}_{L1} \bar{e}_{L2} \bar{e}_{L3}) m_{LR}^E \begin{pmatrix} e_{R1} \\ e_{R2} \\ e_{R3} \end{pmatrix} - \frac{1}{2} (\bar{\nu}_{L1} \bar{\nu}_{L2} \bar{\nu}_{L3}) m_{LL} \begin{pmatrix} \nu_{L1}^c \\ \nu_{L2}^c \\ \nu_{L3}^c \end{pmatrix} + \text{h.c.} \quad (1.1)$$

where e_{Li} represent the three left-handed charged lepton states, e_{Ri} represent the right-handed charged lepton states with ν_{Li} representing the three left-handed neutrino states and ν_{Li}^c are their CP conjugates.

It must be kept in mind that ν_{Li} are not the mass eigenstate neutrinos since m_{LL} is not diagonal in general. We shall refer to the mass eigenstate neutrinos as ν_i .

Neutrino and charged lepton masses are given by the eigenvalues of a complex charged lepton mass matrix, m_{LR}^E , and complex symmetric neutrino Majorana matrix, m_{LL} , obtained by diagonalizing these mass matrices,

$$V^{E_L} m_{LR}^E V^{E_R \dagger} = \begin{pmatrix} m_e & 0 & 0 \\ 0 & m_\mu & 0 \\ 0 & 0 & m_\tau \end{pmatrix}, \quad (1.2)$$

$$V^{\nu_L} m_{LL} V^{\nu_L T} = \begin{pmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{pmatrix}. \quad (1.3)$$

where V^{E_L} , V^{E_R} and V^{ν_L} are unitary transformations on the left-handed charged lepton fields ' E_L ', right-handed charged lepton fields ' E_R ' and left-handed neutrino fields, ν_L , which put the mass matrices into diagonal form with real eigenvalues.

We can then construct the lepton mixing matrix as $U = V^{E_L} V^{\nu_L \dagger}$. Hence neutrino mixing matrix,

$$m_{LL} = U \begin{pmatrix} m_1 & & \\ & m_2 & \\ & & m_3 \end{pmatrix} U^\dagger \quad (1.4)$$

which is something many neutrino physicists are familiar with.

1.2 Heirarchy and Types of SeeSaw Mechanisms

Since, we now have 3 neutrino masses, the experimental constraints put them into 3 possible heirarchy structures¹;

- **Normal Hierarchy:** $m_1^2 \sim m_2^2 \ll m_3^2$
- **Inverted Hierarchy:** $m_1^2 \sim m_2^2 \gg m_3^2$
- **Approximate Degeneracy:** $m_1^2 \sim m_2^2 \sim m_3^2$

	Type I Small $\beta\beta_{0\nu}$	Type II Large $\beta\beta_{0\nu}$
A		
Normal hierarchy	$\beta\beta_{0\nu} \lesssim 0.0082 \text{ eV}$	
$m_1^2, m_2^2 \ll m_3^2$	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \frac{m}{2}$	—
B		
Inverted hierarchy	$\beta\beta_{0\nu} \lesssim 0.0082 \text{ eV}$	$\beta\beta_{0\nu} \gtrsim 0.0085 \text{ eV}$
$m_1^2 \approx m_2^2 \gg m_3^2$	$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \frac{m}{\sqrt{2}}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} m$
C		
Approximate degeneracy		$\beta\beta_{0\nu} \gtrsim 0.035 \text{ eV}$ $\text{diag}(1, 1, 1)m$
$m_1^2 \approx m_2^2 \approx m_3^2$	$\begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} m$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} m$

Table 1.1: **Table 1.** Leading order low energy neutrino Majorana mass matrices, m_{LL} , classified according to the rate of neutrinoless double beta decay and the pattern of neutrino masses.

¹third not so much but for some variety and as a case let's look at that.

where we have only shown the leading order forms explicitly in table above. As we can see, we have classified the mass matrices into 2 parts i.e. **Type I: small neutrinoless beta decay** and **Type II: Large neutrinoless beta decay** other than our usual mass hierarchy classification.

$$\beta\beta_{0\nu} = \Sigma_i U_{ei}^2 m_i \quad (1.5)$$

We can see that there is only one neutrino mass matrix consistent with the normal neutrino mass hierarchy, which we call Type IA (leading order neutrino masses of form $m_i = (0, 0, m)$).

For the inverted hierarchy, there are two cases, Type IB ($m_i = (m, -m, 0)$), and Type IIB ($m_i = (m, m, 0)$).

For the approximate degeneracy cases, there are three cases, Type IC ($m_i = (m, -m, m)$), and two Type IIC (i.e. $m_i = (m, m, m)$ or $m_i = (m, m, -m)$).

1.3 SeeSaw Mechanism

We shall be focusing more on Type-I SeeSaw and some on Type-II mechanism.

1.3.1 Type-I SeeSaw

We have neutrino majorana mass term of form, $m_{LL}\bar{\nu}_L\nu_L^c$, where ν_L is a left-handed neutrino field and ν_L^c is CP conjugate of left-handed neutrino field, i.e. a right-handed antineutrino field. This term is possible since neutrinos do not interact with EM. Still they are **Gauge variant**, thus we assume (in the simplest version of the see-saw mechanisms) that such terms are zero to begin with (in SM) but are generated effectively after right-handed neutrinos are introduced. By introducing right handed fields, we get 2 types of additional neutrino mass terms. Additional **majorana masses** are of form,

$$M_{RR}\bar{\nu}_R\nu_R^c \quad (1.6)$$

where ν_R is a right-handed neutrino field and ν_R^c is the CP conjugate of a right-handed neutrino field, i.e. a left-handed antineutrino field. Other than that there are **Dirac Masses** of form,

$$m_{LR}\bar{\nu}_L\nu_R \quad (1.7)$$

which conserve lepton numbers as well as consistent with electric charge conservation. Hence we can form a mass matrix²,

$$\begin{pmatrix} \bar{\nu}_L & \bar{\nu}_R^c \end{pmatrix} \begin{pmatrix} 0 & m_{LR} \\ m_{LR}^T & M_{RR} \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \end{pmatrix}. \quad (1.8)$$

The right-handed neutrinos are electroweak singlets, thus the Majorana masses of the right-handed neutrinos, M_{RR} , may be orders of magnitude larger than the electroweak

²Assuming no Higgs triplet

scale. For the approximation $M_{RR} \gg m_{LR}$, then we can diagonalize the matrix above to get effective Majorana masses of the type listed before. For that case,

$$m_{LL} = -m_{LR}M_{RR}^{-1}m_{LR}^T \quad (1.9)$$

The effective left-handed Majorana masses, m_{LL} , are naturally suppressed by the heavy scale M_{RR} . One may derive the See-Saw Mechanism from Lagrangian given;

$$\mathcal{L} = -\bar{\nu}_L m_{LR} \nu_R - \frac{1}{2} \nu_R^T M_{RR} \nu_R + \text{H.c.}, \quad (1.10)$$

where ν_L represents left-handed neutrino fields (from electroweak doublets) and ν_R represents right-handed neutrino fields (from electroweak singlets), in a matrix notation where the m_{LR} matrix elements are typically of order the charged lepton masses, while the M_{RR} matrix elements may be much larger than the electroweak scale³. The number of right-handed neutrinos is not fixed, but the number of left handed is fixed as 3. Below the mass scale of the right-handed neutrinos, we can integrate them out using the equations of motion

$$\frac{d\mathcal{L}}{d\nu_R} = 0, \quad (1.11)$$

giving us

$$\nu_R^T = -\bar{\nu}_L m_{LR} M_{RR}^{-1}, \quad \nu_R = -M_{RR}^{-1} m_{LR}^T \bar{\nu}_L^T. \quad (1.12)$$

Substituting it into original Lagrangian, we get

$$\mathcal{L} = -\frac{1}{2} \bar{\nu}_L m_{LL} \nu_L^c + \text{H.c.} \quad (1.13)$$

where, $m_{LL} = -m_{LR}M_{RR}^{-1}m_{LR}^T$.

This is the the Type I see-saw mechanism, the simplest version that can be thought of as resulting from integrating out heavy right-handed neutrinos to produce the effective dimension 5 neutrino mass operator

$$-\frac{1}{2} H_u L^T \kappa H_u L,$$

where

$$\kappa = Y_{LR}^\nu M_{RR}^{-1} Y_{LR}^{\nu T}.$$

³may be up to the Planck scale

1.3.2 Type II SeeSaw

One might think of writing a similar operator by hand. One such example would be the operator by Weinberg which was, suppressed by the Planck scale (rather than the right-handed neutrino mass scales unlike Type I). It was introduced in order to account for small neutrino masses only but such a Planck scale suppressed operator led to neutrino masses of the order of 10^{-5} eV, too small to account for m_2 or m_3 (though they could account for m_1 [N.O.]⁴). To account for m_3 [in N.O.], we need a dimension 5 operators suppressed by a mass scale of order $3 * 10^{14}$ GeV given that the dimensionless coupling of the operator is of order '1' and the Higgs vev is equal to that of Standard Model.

We can also generate dimension 5 operator by exchange of heavy Higgs triplets of $SU(2)_L$ only, which we refer to as the Type II see-saw mechanism.

⁴Opposite for I.O.

Chapter 2

Deviation II-SUSY[9]

One useful deviation would be to study Supersymmetry and Minimal SuperSymmetric Standard Model(MSSM), but such a deviation would be too much for current project which aims at other things. Nevertheless we may study it later and the report shall be added here to complete this report. We might follow[9] for the same.¹

¹Can be done in course report for PHY639

Chapter 3

Scotogenic model[1]

Scotogenic Model of Neutrino Mass Mechanism and Dark Matter is probably one of the easiest to get into, where left handed neutrinos obtain naturally small Majorana masses from one-loop radiative seesaw mechanism. In Dark matter, it proposed 2 candidates, one bosonic and one fermionic.

For a fully theoretical understanding of the model, some more things than the aforementioned is required, mainly,

- **Higgs Doublet Models:** 1106.0034,0603188,0612275
- **Radiative Neutrino Mass Models:** 1706.08524,1308.3655
- **Others:** 9805219,1312.2840

But such a detailed understanding is not required at this point of project and we will take forward from the chapter on SeeSaw Model. These theoretical aspects are important for a full picture so we will tackle them sometime later.

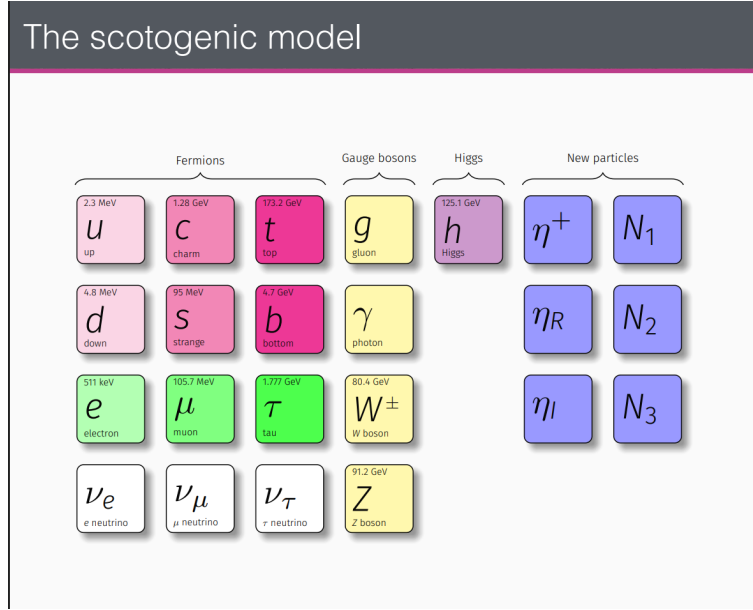


Figure 3.1: SM under Scotogenic model. Taken from slide of Ivania Maturana Ávila, IFIC[10]

3.1 Naturally Small Neutrino Masses[11]

For the minimal standard electroweak gauge model, there exist an effective dimension-five operator which generates neutrino masses as discussed which has only three tree-level realizations with one being the canonical seesaw mechanism with a right-handed neutrino. Another being a heavy Higgs triplet¹. The third is to having a heavy Majorana fermion triplet, an example of which is presented here in the context of supersymmetric SU(5) grand unification.

For the minimal Standard Model of quarks and leptons, each of the three neutrinos appears only as a member of a left-handed SU(2) lepton doublet of type $\psi = (\nu_i, l_i)_L$ like in SM. However the Higgs appears as a scalar doublet, $\Phi = (\phi^+, \phi^0)$ leading to massless neutrinos.

Due to the effective dimension-five operator in minimal standard electroweak gauge model, which generates majorana masses, $\Lambda^{-1} \phi^0 \phi^0 \nu_i \nu_j$ with Λ as large effective mass. In order to obtain the effective operator at tree level, using only 'renormalizable' interactions, there can be three ways:²

- ψ_i and Φ form a fermion singlet:

$$\Lambda^{-1} (\phi^0 \nu_i - \phi^+ l_i) (\phi^0 \nu_j - \phi^+ l_j) \quad (3.1)$$

¹Some Interesting work done by Rohini Godbole

²Using simple power counting

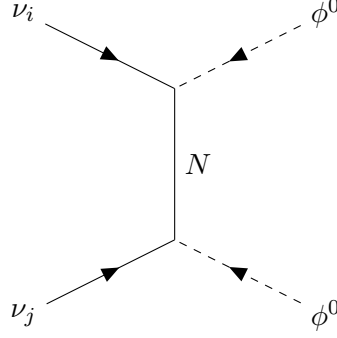


Figure 3.2: Tree-level realization of the effective 5-D operator with fermion singlet

The intermediate heavy particle in this case is a fermion singlet (Say N , with mass M and coupling to ν_i as f_i). Then obviously $\lambda^{-1} = f_i f_j / 2M$ as we already did while extending Fermi Theory to iVB theory. Since ϕ^0 has a non zero vev, $f_i \nu$ can be seen as a Dirac mass m_i , linking n_i to N and the neutrino mass matrix simply becomes $-m_i m_j / M$. This is our SeeSaw mechanism with N being right handed neutrino with a large Majorana mass. Ofcourse we need 3 N 's for a usual scenario.

- ψ_i and ψ_j form a scalar triplet:³

$$\Lambda^{-1} [\phi^0 \phi^0 \nu_i \nu_j - \phi^+ \phi^0 (\nu_i l_j + l_i \nu_j) + \phi^+ \phi^+ l_i l_j], \quad (3.2)$$

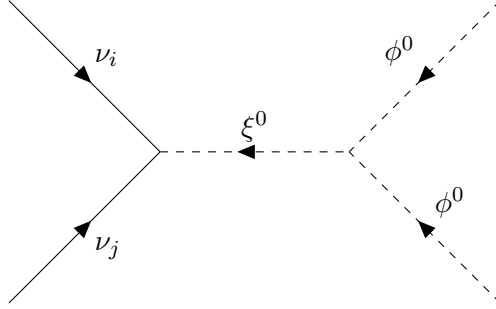


Figure 3.3: Tree-level realization of the effective 5-D operator with heavy scalar triplet.

We need a heavy scalar triplet $\xi \equiv (\xi^{++}, \xi^+, \xi^0)$ (say mass, M and coupling to $\nu_i \nu_j$ and $\phi^0 \phi^0$ are f_{ij} and μ resp.) then $\Lambda^{-1} = f_{ij} \mu / M^2$ and neutrino mass matrix is given by,

$$(\mathcal{M}_\nu)_{ij} = -2 f_{ij} \mu \nu^2 / M^2 \quad (3.3)$$

³Singlet combination of ψ_i, ψ_j is $\nu_i l_j - l_i \nu_j$ does not formulate $\Lambda^{-1} \phi^0 \phi^0 \nu_i \nu_j$

only one ξ is required for all neutrinos to become massive, thus not requiring right-handed neutrinos. This theory falls down to the singlet model in low Energy limit as one can easily see. Even though ξ is heavy, it has a tiny vev of about $u = -\mu\nu^2/M^2$, the neutrino mass matrix being equal to $2f_{ij}u$ as expected from the direct coupling of ξ to $\nu_i\nu_j$. Due to this it is not largely appreciated.

- ψ_i and Φ form a fermion triplet:

$$\Lambda^{-1}[(\phi^0\nu_i + \phi^+l_i)(\phi^0\nu_j + \phi^+l_j) - 2\phi^+\nu_i\phi^0l_j - 2\phi^0l_i\phi^+\nu_j]. \quad (3.4)$$

Instead of N as in the singlet model, we replace it by a heavy majorana fermionic triplet $\Sigma \equiv (\Sigma^+, \Sigma^0, \Sigma^-)$ leading to a seesaw. It is similar to the previous 2 model in low energy limit as expected.

In the singlet model, addition of three N's argues favorably for SO(10) instead of SU(5) as a suitable unifying symmetry, whereas in triplet models SU(5) is sufficient. In particular, the 15 group representation SU(5) would contain ξ , while 24 representation would contain both N and Σ .

Eg. we can break SU(5) to the standard $SU(3) \otimes SU(2) \otimes U(1)$ using the 24 supermultiplet, i.e. $24 = (1, 1, 0) + (8, 1, 0) + (1, 3, 0) + (3, 2, -5/6) + (3^*, 2, 5/6)$, the scalar component of $(1, 1, 0)$ acquires a large vev, fermionic components of $(1, 1, 0)$ and $(1, 3, 0)$ will be N and Σ only.

3.2 Scotogenic Model

⁴ We add 3 heavy singlet majorana particle neutrinos N_i ($i = \{1, 2, 3\}$) to SM such that,

$$\mathcal{M}_\nu^{(e, \mu, \tau)} = -\mathcal{M}_D \mathcal{M}_N^{-1} \mathcal{M}_D^T \quad (3.5)$$

where $\mathcal{M}_D$ is 3x3 dirac mass matrix linking observed neutrinos to N_i and $\mathcal{M}_N$ is majorana mass matrix of N_i with $\mathcal{M}_\nu$ coming out of unique 5-D operator,⁵

$$\mathcal{L}_\Lambda = \frac{f_{ij}}{\Lambda} (\nu_i \phi^0 - l_i \phi^+) (\nu_j \phi^0 - l_j \phi^+) + H.c. \quad (3.6)$$

We have (ν_i, l_i) as the usual left-handed lepton doublets transforming as $(2, -1/2)$ ⁶ under standard Electroweak gauge group, $SU(2)_L \otimes U(1)_Y$ whereas, $(\phi^+, \phi^0) \sim (2, 1/2)$ is the usual Higgs doublet of the SM. As discussed in last section, this has 3 generic mechanism to get this operator in one loop. The new particles required in the three tree-level realizations are most likely too heavy to be observed experimentally in near future but

⁴Interestingly the name is derived from greek with skotos ($\sigma\kappa\omicron\tau\omicron\varsigma$), meaning "darkness," and genesis ($\gamma\epsilon\upsilon\epsilon\sigma\iota\varsigma$), meaning "origin" or "creation." Coined by Ernest Ma in 2006, it literally translates to "generated by darkness" or "born of dark matter."

⁵Remember Seesaw mech chapter and Singlet model in previous section

⁶(2 dim rep of SU(2) here, hypercharge)

those involved in the one-loop realizations may be light enough to be detected, in LHC etc.

Considering minimal extension of SM under $SU(2)_L \otimes U(1)_Y \otimes Z_2$, then the particle content will be,

$$\begin{aligned} (\nu_i, l_i) &\sim (2, -1/2; +), \quad l_i^c \sim (1, 1; +), \quad N_i \sim (1, 0; -), \\ (\phi^+, \phi^0) &\sim (2, 1/2; +), \quad (\eta^+, \eta^0) \sim (2, 1/2; -). \end{aligned}$$

The tree and loop level mechanisms of the previous section already show that a $\mathbb{Z}_2$ -odd sector coupled to the lepton doublet through a second scalar doublet is one of the generic ways of realising the dimension-five (Weinberg) operator [12] radiatively rather than at tree level [11]. Ma's 2006 paper [1] makes this loop realisation completely explicit and, crucially, promotes the discrete symmetry that forbids the tree-level piece to an exact symmetry of full Lagrangian, such that the loop diagram becomes the only source of neutrino mass. This change from a softly broken global lepton number [13] to an exact $\mathbb{Z}_2$ is what makes the model simultaneously natural, since $\lambda_5 \rightarrow 0$ restores lepton number and the neutrinos become exactly massless, and predictive of dark matter, since the same $\mathbb{Z}_2$ that protects the loop also stabilises the lightest odd particle.

3.2.1 Field content and the exact $\mathbb{Z}_2$ symmetry

In addition to the Standard Model lepton doublets $(\nu_i, l_i)_L$, right-handed charged leptons l_i^c , and the Higgs doublet $\Phi = (\phi^+, \phi^0)$, the model adds three neutral singlet Majorana fermions N_i ($i = 1, 2, 3$) and one extra scalar doublet $\eta = (\eta^+, \eta^0)$ carrying the same gauge quantum numbers as Φ . An extra discrete symmetry $\mathbb{Z}_2$ is imposed, under which N_i and η are odd and every Standard Model field is even⁷, Table 3.1.

⁷Since $\mathbb{Z}_2$ is the simplest group with only 2 elements, we rep them by + and -, or even and odd respectively. This is a multiplicative Quantum Number. Every Standard Model field(here) is even, and only the two new pieces added by the model (N and η) are odd. Also to make sure the sign doesn't flip every term in the Lagrangian must be even, hence, only even number of odd fields are allowed in lagrangian as we will see. This has a few consequences like

- η^0 can't get a vacuum expectation value. Since, vev is "the value the field settles to." If an odd field got a nonzero vev, the vacuum itself would prefer a specific sign, which spontaneously breaks $\mathbb{Z}_2$. So keeping its vev 0 keeps symmetry exact
- No tree-level neutrino mass. As shown above, we simply can't write a $\nu - N$ Dirac mass term without violating the odd-count rule, so neutrino mass is zero unless we go to loop level, where two odd particles (η and N) run around the loop and the diagram has an even number of them.
- The lightest odd particle is absolutely stable which for us is the dark matter. Any decay of an odd particle has to end in a final state that's still net-odd (since the initial state was odd, and $\mathbb{Z}_2$ -charge is conserved). So the lightest odd particle has nothing lighter that's also odd to decay into... i.e. it's stuck, forever.

Field	$SU(2)_L$	$U(1)_Y$	$\mathbb{Z}_2$
$(\nu_i, l_i)_L$	2	$-1/2$	+
l_i^c	1	1	+
N_i	1	0	−
(ϕ^+, ϕ^0)	2	$1/2$	+
(η^+, η^0)	2	$1/2$	−

Table 3.1: Field content of the scotogenic model and its $SU(2)_L \times U(1)_Y \times \mathbb{Z}_2$ charges, $i = 1, 2, 3$ [1].

Because $\mathbb{Z}_2$ is exact and not softly broken, unlike in the closely related model of [13], where the same field content is used but the discrete symmetry is replaced by a global lepton number broken explicitly, if softly, by a bilinear $\mu^2 \Phi^\dagger \eta + H.c.$ term. Term of this kind is not allowed in the present model as reason mentioned in previous footnote.

Also the lightest $\mathbb{Z}_2$ -odd particle (LOP) either the lightest N_k , or the lighter of the two neutral scalars η_R, η_I defined above is absolutely stable, because every vertex of the theory contains an even number of $\mathbb{Z}_2$ -odd legs. At some places it has been mentioned as analogue of R -parity in the MSSM, and turns our neutrino-mass model into a dark-matter model freely.

3.2.2 Yukawa interactions and the scalar potential

The Yukawa Lagrangian of the model is

$$\mathcal{L}_Y = f_{ij} (\phi^- \nu_i + \bar{\phi}^0 l_i) l_j^c + h_{ij} (\nu_i \eta^0 - l_i \eta^+) N_j + H.c., \quad (3.7)$$

together with the Majorana mass term $\frac{1}{2} M_i N_i N_i + H.c.$ for the singlets⁸. The first term of Eq. (3.7) is the usual charged-lepton Yukawa coupling; the second couples the lepton doublet to N_j through the inert doublet η and is the only new Yukawa structure introduced by the model. No term of the form $\bar{\nu}_i N_j \phi^0$ is allowed, precisely because N_j is $\mathbb{Z}_2$ -odd while ϕ^0 is even.

The most general renormalisable, gauge- and $\mathbb{Z}_2$ -invariant scalar potential for Φ and η is

$$V = m_1^2 \Phi^\dagger \Phi + m_2^2 \eta^\dagger \eta + \frac{1}{2} \lambda_1 (\Phi^\dagger \Phi)^2 + \frac{1}{2} \lambda_2 (\eta^\dagger \eta)^2 + \lambda_3 (\Phi^\dagger \Phi) (\eta^\dagger \eta) + \lambda_4 (\Phi^\dagger \eta) (\eta^\dagger \Phi) + \frac{1}{2} \lambda_5 \left[(\Phi^\dagger \eta)^2 + H.c. \right], \quad (3.8)$$

⁸Gauge invariance under $SU(2)_L$ fixes the index structure of the second term of Eq. (3.7) uniquely: ν_i and l_i sit inside the *same* doublet $(\nu_i, l_i)_L$ and must therefore carry the same generation index, exactly as they do in the f_{ij} term, with the second (Yukawa-matrix) index carried by N_j alone.

where λ_5 can be chosen real by rephasing η appropriately. The bilinear $\mu^2 \Phi^\dagger \eta$ allowed in [13] is absent here, since $\mathbb{Z}_2$ forbids it; this is the *only* structural difference between the two potentials, and is responsible for turning a model with softly broken lepton number into one with a stable dark-matter candidate.

3.2.3 Electroweak symmetry breaking and the scalar mass spectrum

Vacuum and tadpole condition

For $m_1^2 < 0$ and $m_2^2 > 0$, only ϕ^0 acquires a vacuum expectation value. Thus,

$$\phi^0 = v + \frac{1}{\sqrt{2}}(H + i\chi), \quad \eta^0 = \frac{1}{\sqrt{2}}(\eta_R + i\eta_I), \quad (3.9)$$

with $v \equiv \langle \phi^0 \rangle$ real and $\phi^+ = \eta^+$ carrying no vev. (We have followed this convention throughout in order to derive the quoted in [1] exactly. Numerically $v \simeq 174$ GeV, i.e. $v = v_{\text{SM}}/\sqrt{2}$, $v_{\text{SM}} = 246$ GeV) H is the physical Higgs boson and χ the neutral Goldstone mode is absorbed by the Z .

Using $\Phi^\dagger \Phi = (v + H/\sqrt{2})^2 + \chi^2/2$ and dropping η (which carries no vev and so does not affect the tadpole condition), the H -dependent part of the potential is $V(H) = m_1^2 (v + H/\sqrt{2})^2 + \frac{\lambda_1}{2} (v + H/\sqrt{2})^4$. Demanding $\partial V/\partial H|_{H=0} = 0$ gives the minimisation condition

$$m_1^2 = -\lambda_1 v^2, \quad (3.10)$$

identical to the pure Standard-Model relation. Expanding $V(H)$ to $O(H^2)$ and substituting this back in,

$$V(H) = \text{const} + (m_1^2 + 3\lambda_1 v^2) \frac{H^2}{2} + O(H^3) = \text{const} + \lambda_1 v^2 H^2 + O(H^3), \quad (3.11)$$

so that, matching $V \supset \frac{1}{2} m_H^2 H^2$,

$$m^2(\sqrt{2} \text{Re } \phi^0) \equiv m_H^2 = 2\lambda_1 v^2, \quad (3.12)$$

which forms Eq. (7) of [1]. It seems to be simply ordinary Standard-Model Higgs mass formula, unaffected at tree level by the inert sector.

The inert-doublet spectrum

Since η carries no vev, its mass matrix follows directly from the η -bilinear terms of V evaluated at $\Phi \rightarrow \begin{pmatrix} 0 \\ v \end{pmatrix}$, using $\Phi^\dagger \eta \rightarrow v\eta^0$ and $\eta^\dagger \Phi \rightarrow v\eta^{0*}$, we get,

$$V|_{\eta\text{-bilinear}} = (m_2^2 + \lambda_3 v^2) \eta^+ \eta^- + (m_2^2 + \lambda_3 v^2 + \lambda_4 v^2) |\eta^0|^2 + \frac{\lambda_5 v^2}{2} [(\eta^0)^2 + (\eta^{0*})^2]. \quad (3.13)$$

The charged component does not mix with anything, so its mass is read off immediately,

$$m^2(\eta^\pm) = m_2^2 + \lambda_3 v^2, \quad (3.14)$$

producing Eq. (8) of [1]. For the neutral component, $|\eta^0|^2 = \frac{1}{2}(\eta_R^2 + \eta_I^2)$ and $(\eta^0)^2 + (\eta^{0*})^2 = \eta_R^2 - \eta_I^2$, so

$$V|_{\eta^0\text{-bilinear}} = \frac{1}{2} [m_2^2 + (\lambda_3 + \lambda_4 + \lambda_5)v^2] \eta_R^2 + \frac{1}{2} [m_2^2 + (\lambda_3 + \lambda_4 - \lambda_5)v^2] \eta_I^2 : \quad (3.15)$$

$\eta_R \equiv \sqrt{2}\text{Re}\eta^0$ and $\eta_I \equiv \sqrt{2}\text{Im}\eta^0$ are *already* the mass eigenstates (since, there is no η_R - η_I mixing) and

$$m_R^2 \equiv m^2(\sqrt{2}\text{Re}\eta^0) = m_2^2 + (\lambda_3 + \lambda_4 + \lambda_5)v^2, \quad (3.16)$$

$$m_I^2 \equiv m^2(\sqrt{2}\text{Im}\eta^0) = m_2^2 + (\lambda_3 + \lambda_4 - \lambda_5)v^2, \quad (3.17)$$

i.e. Eqs. (9)–(10) of [1]. The two neutral states are split entirely by λ_5 ,

$$m_R^2 - m_I^2 = 2\lambda_5 v^2, \quad (3.18)$$

the quantity that controls the size of the neutrino mass below. For $\lambda_5 = 0$ the potential Eq. (3.8) conserves a $U(1)$ lepton number under which $\eta \rightarrow e^{i\alpha}\eta$, $N_i \rightarrow e^{-i\alpha}N_i$ (with ν_i, l_i uncharged, so that h_{ij} conserves this $U(1)$), η_R and η_I then become exactly degenerate. The neutrino mass vanishes in this limit too. Since, lepton number is broken only by $\lambda_5 \neq 0$, so the neutrino mass and the mass splitting $m_R^2 - m_I^2$ must be proportional to the same symmetry-breaking parameter.

3.2.4 One-loop generation of the neutrino mass matrix

Because $\langle \eta^0 \rangle = 0$, $\mathcal{M}_\nu$ vanishes at tree level; it is generated at one loop by the diagram of Fig. 3.4, in which a neutrino line converts into N_k , emits and reabsorbs a neutral scalar, and converts back.

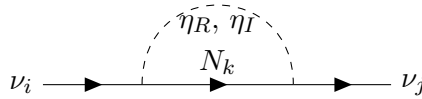


Figure 3.4: One-loop generation of $(\mathcal{M}_\nu)_{ij}$ (Fig. 1 of [1]), through the exchange of the heavy singlet N_k and the neutral scalars η_R, η_I .⁹

Calculation

Writing $\eta^0 = (\eta_R + i\eta_I)/\sqrt{2}$ in Eq. (3.7), the relevant piece of the Yukawa Lagrangian becomes

$$\mathcal{L}_Y \supset \frac{h_{ik}}{\sqrt{2}} \nu_i \eta_R N_k + \frac{ih_{ik}}{\sqrt{2}} \nu_i \eta_I N_k + H.c. \quad (3.19)$$

In four-component language, η_R couples through an ordinary scalar vertex $\propto P_L$, while η_I couples through a pseudoscalar-type vertex $\propto iP_L$. Squaring the amplitude, this relative i becomes a relative factor of $i \times i = -1$ between the η_R - and η_I -mediated diagrams i.e. **the two contributions subtract rather than add**. This sign is the origin of the minus sign in Eq. (3.23) below and, combined with Eq. (3.18), is what forces $(\mathcal{M}_\nu)_{ij} \propto \lambda_5$.

For each real scalar exchanged, the graph is a fermion self-energy diagram built from an internal Majorana fermion N_k (mass M_k) and an internal real scalar (mass $m = m_R$ or m_I). Because N_k is Majorana, the loop closes using the lepton-number-violating (wrong chirality, mass-insertion) piece of its propagator directly, so the piece of the loop integral relevant for the $\bar{\nu}_i^c \nu_j$ (Majorana mass) operator is proportional to M_k times the scalar loop integral

$$J(M_k, m) \equiv \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 - M_k^2)(k^2 - m^2)} = \frac{i}{16\pi^2} \int_0^1 dx \left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi - \ln \Delta(x) \right], \quad (3.20)$$

with $d = 4 - 2\epsilon$ and $\Delta(x) = xM_k^2 + (1-x)m^2$, taken the usual Feynman-parameter combination from $\frac{1}{AB} = \int_0^1 \frac{dx}{[xA + (1-x)B]^2}$. Using

$$g(M^2, m^2) \equiv \int_0^1 \ln [xM^2 + (1-x)m^2] dx = \ln M^2 + \frac{m^2}{M^2 - m^2} \ln \frac{M^2}{m^2} - 1, \quad (3.21)$$

(obtained by direct integration, substituting $u = xM^2 + (1-x)m^2$, $du = (M^2 - m^2)dx$), the finite, mass-dependent part of $J(M_k, m)$ is thus, $-\frac{i}{16\pi^2} g(M_k^2, m^2)$.

Since, the η_R - and η_I -mediated diagrams enter with a relative minus sign but are otherwise identical, so the divergent piece of $J(M_k, m_R)$, which is x -independent, and hence identical to that of $J(M_k, m_I)$ cancels exactly in the difference $J(M_k, m_R) - J(M_k, m_I)$ i.e. **the scotogenic neutrino mass is completely UV finite**, with no counterterm required. What survives the cancellation is,

$$g(M_k^2, m_R^2) - g(M_k^2, m_I^2) = \frac{m_R^2}{m_R^2 - M_k^2} \ln \frac{m_R^2}{M_k^2} - \frac{m_I^2}{m_I^2 - M_k^2} \ln \frac{m_I^2}{M_k^2}, \quad (3.22)$$

the $\ln M_k^2$ and (-1) pieces of Eq. (3.21) having cancelled between the η_R and η_I terms. Collecting the Yukawa factors $h_{ik}h_{jk}$, the Majorana mass insertion M_k , and the loop normalisation (the overall coefficient $1/16\pi^2$, rather than the Dirac-loop value $1/32\pi^2$, follows from the extra factor of two associated with the two independent Wick contractions available for a Majorana propagator) and summing over the three singlets we get,

$$(\mathcal{M}_\nu)_{ij} = \sum_k \frac{h_{ik}h_{jk}M_k}{16\pi^2} \left[\frac{m_R^2}{m_R^2 - M_k^2} \ln \frac{m_R^2}{M_k^2} - \frac{m_I^2}{m_I^2 - M_k^2} \ln \frac{m_I^2}{M_k^2} \right], \quad (3.23)$$

which formulates Eq. (11) of [1]. As a check, setting $m_R = m_I$ (i.e. $\lambda_5 = 0$) makes the bracket vanish identically, since both terms become same function of M_k^2 once $m_0^2 \equiv m_R^2 = m_I^2$. this exact cancellation is the loop-level showing of restored lepton-number symmetry, is a good consistency check that Eq. (3.23) needs no separate fine-tuning to protect the neutrino mass.

3.2.5 Limiting cases and the effective seesaw scale

Small mass splitting

If $m_R^2 - m_I^2 = 2\lambda_5 v^2$ is small compared to $m_0^2 \equiv (m_R^2 + m_I^2)/2$, Eq. (3.23) may be Taylor-expanded about m_0^2 . Writing $f(x) \equiv \frac{x}{x - M_k^2} \ln \frac{x}{M_k^2}$, so that the bracket of Eq. (3.23) reads $f(m_R^2) - f(m_I^2) \simeq (m_R^2 - m_I^2)f'(m_0^2)$, using

$$f'(x) = \frac{1}{x - M_k^2} \left[1 - \frac{M_k^2}{x - M_k^2} \ln \frac{x}{M_k^2} \right], \quad (3.24)$$

Eq. (3.23), to leading order in the splitting,

$$(\mathcal{M}_\nu)_{ij} = \frac{\lambda_5 v^2}{8\pi^2} \sum_k \frac{h_{ik} h_{jk} M_k}{m_0^2 - M_k^2} \left[1 - \frac{M_k^2}{m_0^2 - M_k^2} \ln \frac{m_0^2}{M_k^2} \right], \quad (3.25)$$

becomes Eq. (12) of [1]. This step is exact in the limit $2\lambda_5 v^2 \ll m_0^2$.

Heavy singlets, $M_k^2 \gg m_0^2$

Writing $M_k^2 = m_0^2/\varepsilon$ with $\varepsilon \rightarrow 0$, leads to $m_0^2 - M_k^2 \rightarrow -M_k^2$ and $\ln(m_0^2/M_k^2) \rightarrow -\ln(M_k^2/m_0^2)$, so the bracket of Eq. (3.25) tends to $1 - \ln(M_k^2/m_0^2)$ and $M_k/(m_0^2 - M_k^2) \rightarrow -1/M_k$, giving

$$(\mathcal{M}_\nu)_{ij} = \frac{\lambda_5 v^2}{8\pi^2} \sum_k \frac{h_{ik} h_{jk}}{M_k} \left[\ln \frac{M_k^2}{m_0^2} - 1 \right], \quad (3.26)$$

i.e. Eq. (13) of [1], as in the canonical seesaw, the mass is suppressed by $1/M_k$, up to the logarithm and explicit prefactor $\lambda_5 v^2/8\pi^2$.

Light singlets, $m_0^2 \gg M_k^2$

Here $m_0^2 - M_k^2 \rightarrow m_0^2$, and the logarithmic term of Eq. (3.25) is suppressed by an extra power of $M_k^2/m_0^2 \rightarrow 0$ relative to the leading “1”, so the bracket term $\rightarrow 1$ and

$$(\mathcal{M}_\nu)_{ij} = \frac{\lambda_5 v^2}{8\pi^2 m_0^2} \sum_k h_{ik} h_{jk} M_k, \quad (3.27)$$

is Eq. (14) of [1] i.e. the neutrino mass now grows with M_k rather than being suppressed by it, since the loop is instead dominated by the light scalar propagators.

Near-degenerate singlet and scalar masses, $m_0^2 \simeq M_k^2$

Let's set $M_k^2 = m_0^2(1 - \varepsilon)$ and expand to $O(\varepsilon)$. One gets,

$$\ln \frac{m_0^2}{M_k^2} = -\ln(1 - \varepsilon) = \varepsilon + \frac{\varepsilon^2}{2} + O(\varepsilon^3), \quad \frac{M_k^2}{m_0^2 - M_k^2} = \frac{1 - \varepsilon}{\varepsilon}, \quad (3.28)$$

Hence, the bracket term of Eq. (3.25) becomes

$$1 - \frac{1 - \varepsilon}{\varepsilon} \left(\varepsilon + \frac{\varepsilon^2}{2} + O(\varepsilon^3) \right) = 1 - (1 - \varepsilon) \left(1 + \frac{\varepsilon}{2} + O(\varepsilon^2) \right) = \frac{\varepsilon}{2} + O(\varepsilon^2), \quad (3.29)$$

(the $O(\varepsilon^0)$ and $O(\varepsilon^1)$ terms cancelling between the two factors), while the prefactor $M_k/(m_0^2 - M_k^2) = M_k/(m_0^2 \varepsilon) \rightarrow 1/(m_0 \varepsilon)$ using $M_k \simeq m_0$. The product of the (divergent) prefactor and the (vanishing) bracket is therefore finite, $\rightarrow 1/(2m_0)$, and Eq. (3.25) collapses to

$$(\mathcal{M}_\nu)_{ij} \simeq \frac{\lambda_5 v^2}{16\pi^2} \sum_k \frac{h_{ik} h_{jk}}{M_k}, \quad (3.30)$$

is Eq. (15) of [1] (using $M_k \simeq m_0$ in the last step),¹⁰

Seesaw scale

Compared with the canonical type-I seesaw formula $m_\nu \sim h^2 v^2 / M$ of the previous chapter [7], [14], [15], [16], the scotogenic formula carries an extra explicit factor $\lambda_5 / 16\pi^2$. Taking, as in [1], $h^2 \sim \lambda_5 \sim 10^{-4}$, $v \simeq 174$ GeV and $m_\nu \sim 1$ eV = 10^{-9} GeV, the canonical scale required is

$$M_{\text{canonical}} \sim \frac{h^2 v^2}{m_\nu} = \frac{10^{-4} \times (174 \text{ GeV})^2}{10^{-9} \text{ GeV}} \approx 3 \times 10^9 \text{ GeV}, \quad (3.31)$$

matching the “ 10^9 GeV” quoted in [1]. From Eq. (3.30), the scotogenic scale needed for the *same* m_ν is instead

$$M \sim \frac{\lambda_5 h^2 v^2}{16\pi^2 m_\nu} = \frac{10^{-4} \times 10^{-4} \times (174 \text{ GeV})^2}{16\pi^2 \times 10^{-9} \text{ GeV}} \approx 1.9 \times 10^3 \text{ GeV} \approx 1 \text{ TeV}, \quad (3.32)$$

as told in Ma's paper. up to the unspecified $O(1)$ factors absorbed into “ $h^2 \sim \lambda_5 \sim 10^{-4}$ ”. The ratio of the two scales,

$$\frac{M_{\text{canonical}}}{M} = \frac{16\pi^2}{\lambda_5} \approx 1.6 \times 10^6 \quad (\lambda_5 = 10^{-4}), \quad (3.33)$$

is exactly “factor $\lambda_5 / 16\pi^2$ ” reduction shown in the paper. A loop factor $1/16\pi^2 \approx 6 \times 10^{-3}$ combined with a mildly small coupling $\lambda_5 = 10^{-4}$ is enough to pull the seesaw scale down from a GUT-adjacent value to the electroweak/TeV range, which is exactly what makes the new particles of Table 3.1 plausible LHC targets.

¹⁰this formula is most often quoted for order-of-magnitude estimates, having the simplest, canonical-seesaw-like form.

3.2.6 Dark matter candidates

The exact $\mathbb{Z}_2$ symmetry guarantees a stable lightest odd particle (LOP). The model offers two structurally different candidates depending on the ordering of the N_k and η masses.

- **Fermionic dark matter** (N_1). If $M_1 < M_2, M_3$ and M_1 is below $m^2(\eta^\pm), m_R, m_I$, the lightest singlet N_1 is stable. Its only interactions are the Yukawa couplings h_{i1} of Eq. (3.7), so N_1 annihilates and scatters off nuclei only through t -channel $\eta^\pm$ or $\eta_{R,I}$ exchange. There is no tree-level coupling to the Z or Higgs boson which suppresses direct-detection rates well below current sensitivities for typical Yukawa couplings.¹¹
- **Scalar dark matter**, $\sqrt{2} \text{Re } \eta^0$. If instead the lightest $\mathbb{Z}_2$ -odd state is the lighter neutral scalar (say η_R , with $m_R < m_I, m^2(\eta^\pm), M_k$), it behaves as an inert-doublet-type scalar WIMP of the general kind classified in [18], [19]. Equations (3.16)–(3.17) guarantee that its would-be degenerate partner η_I is automatically only slightly heavier, by $m_I^2 - m_R^2 = -2\lambda_5 v^2$, controlled by the same small coupling λ_5 that also sets the neutrino mass scale. A small splitting between the dark-matter particle and the next-to-lightest odd state is exactly the condition needed for efficient *coannihilation* [20] in the early Universe, and here it follows automatically from the structure of the potential, rather than having to be imposed as an accidental near-degeneracy between two otherwise unrelated superpartners, as is typically the case in supersymmetric dark-matter scenarios[21].

In either case the present-day relic abundance is set by thermal freeze-out, governed by the standard Boltzmann equation for the LOP number density n ,

$$\frac{dn}{dt} + 3Hn = -\langle\sigma v\rangle (n^2 - n_{\text{eq}}^2), \quad (3.34)$$

whose solution gives, parametrically, $\Omega h^2 \propto 1/\langle\sigma v\rangle$.

3.2.7 Collider signatures and decay chains

If the N_k are all lighter than the η states, the observable decay chain is

$$\eta^\pm \rightarrow l^\pm N_{1,2,3}, \quad N_2 \rightarrow l^\pm l^\mp N_1, \quad N_3 \rightarrow l^\pm l^\mp N_{1,2}, \quad (3.35)$$

Eqs. (16)–(18) of [1], the last two proceeding through an off-shell $\eta^\pm$. If instead η is lighter, the chain runs the other way, $N_{1,2,3} \rightarrow l^\pm \eta^\mp$ followed by $\eta^\mp \rightarrow \eta^0 + \{W^\mp\}$, Eqs. (19)–(20)

¹¹Interesting read: neutrino mass is generated at three loops with two extra charged scalar singlets rather than one loop with an inert doublet, was proposed by Krauss, Nasri and Trodden [17] which is a closely related, earlier idea, in which the lightest new fermion is likewise stabilised by a $\mathbb{Z}_2$. The scotogenic model is the minimal one-loop realisation of the same underlying philosophy.

of [1]. In either case, measuring enough of these branching ratios in principle allows the Yukawa matrix h_{ij} to be reconstructed and checked against the neutrino oscillation data through Eq. (3.23).

The two-body width $\Gamma(\eta^+ \rightarrow l_i^+ N_k)$

As an explicit example of the rates entering these chains, consider $\eta^+ \rightarrow l_i^+ N_k$, mediated by vertex $-h_{ik} l_i \eta^+ N_k + H.c.$ of Eq. (3.7), with the charged lepton treated as massless compared with $m_{\eta^+} \equiv m_\eta$ and M_k . Squaring and summing over spins we get,

$$\sum_{\text{spins}} |\mathcal{M}|^2 = |h_{ik}|^2 \text{Tr}[(\gamma \cdot p_l) P_L (\gamma \cdot p_N + M_k) P_R] = 2|h_{ik}|^2 p_l \cdot p_N = |h_{ik}|^2 (m_\eta^2 - M_k^2), \quad (3.36)$$

using $P_L(\gamma \cdot p_N + M_k)P_R = (\gamma \cdot p_N)P_R$ (the mass term drops since $P_L P_R = 0$) and $p_l \cdot p_N = \frac{1}{2}(m_\eta^2 - M_k^2)$, from $m_\eta^2 = (p_l + p_N)^2 = M_k^2 + 2p_l \cdot p_N$ with $p_l^2 = 0$. Using the standard two-body formula $\Gamma = |\vec{p}| |\mathcal{M}|^2 / (8\pi m_\eta^2)$ and $|\vec{p}| = (m_\eta^2 - M_k^2) / 2m_\eta$ for one massless and one massive daughter,

$$\Gamma(\eta^+ \rightarrow l_i^+ N_k) = \frac{|h_{ik}|^2}{16\pi} \frac{(m_\eta^2 - M_k^2)^2}{m_\eta^3}. \quad (3.37)$$

We can see that it vanishes as $M_k \rightarrow m_\eta$ (phase space closes) and reduces to the familiar $\Gamma \rightarrow |h_{ik}|^2 m_\eta / 16\pi$ for $M_k \ll m_\eta$.

Scaling of the three-body width $\Gamma(N_2 \rightarrow l^\pm l^\mp N_1)$

The three-body decays of Eqs. (17)–(18) proceed through an **off-shell** $\eta^\pm$; for $m_\eta \gg M_2$ one may integrate it out and match onto the four-fermion contact operator

$$\mathcal{L}_{\text{eff}} = -\frac{h_{i2} h_{j1}^*}{m_\eta^2} (\bar{l}_i P_L N_2) (\bar{N}_1 P_R l_j) + H.c., \quad (3.38)$$

like we did for case of Fermi contact interaction obtained by integrating out the W boson in muon decay. Repeating the muon-decay phase-space integral (with Dalitz variables $m_{12}^2 = (p_1 + p_2)^2$, $m_{23}^2 = (p_2 + p_3)^2$ and $d\Gamma = |\mathcal{M}|^2 dm_{12}^2 dm_{23}^2 / 256\pi^3 M_2^3$) for this scalar, rather than vector, current-current coupling gives, for $M_1 \ll M_2$,

$$\Gamma(N_2 \rightarrow l_i^- l_j^+ N_1) \sim \frac{|h_{i2}|^2 |h_{j1}|^2}{3072\pi^3} \frac{M_2^5}{m_\eta^4}, \quad (3.39)$$

parametrically of the same form as the muon width $\Gamma_\mu = G_F^2 m_\mu^5 / 192\pi^3$ with $G_F \rightarrow h_{i2} h_{j1} / m_\eta^2$, but with a different $O(10)$ numerical coefficient because the mediator here couples through a scalar, rather than $V-A$, current. The scaling $\Gamma \propto h^4 M_2^5 / m_\eta^4$ (rather

than the precise numerical coefficient which is sensitive to the detailed Dirac structure and to Majorana symmetry factors not tracked in this estimate), is our physically important content as it shows why these decays are suppressed by four powers of (TeV-scale) η mass, and why they become observable only because m_η is pulled down to the TeV range by the loop suppression calculated in Section 3.2.4.

3.2.8 Outlook: family symmetry and tri-bimaximal mixing

Because $\mathcal{M}_\nu$ in Eq. (3.23) can be arranged to be approximately proportional either to $\mathcal{M}_N$ itself, Eq. (3.27), or to its inverse in the heavy-singlet limit, Eq. (3.26), exactly as in the canonical seesaw [22], the model is a natural setting for imposing a discrete family symmetry on h_{ij} and M_i that fixes the mixing pattern independently of the overall mass scale. A_4 is one of the most widely used choice [23], [24], combined with a suitable flavon sector it reproduces the tri-bimaximal mixing pattern [25] that seem to be an excellent zeroth-order description of the observed lepton mixing angles (see e.g. [26] for an explicit A_4 construction along these lines). We are not pursuing this direction further there though it might be a good future project to pursue.

Chapter 4

Implementing Scotogenic model[27]

Numerical implementation of Chapter 3's model via SARAH $\rightarrow$ SPheno $\rightarrow$ MicrOmegas $\rightarrow$ MadGraph, following [27], all at their benchmark point **BS1**.

4.1 S.A.R.A.H.

We used S.A.R.A.H. to define our model, parameter, particles etc which carried on-ward to other programs/packages. Though some preliminary results could be derived from SARAH itself, we stuck around for SPheno to get it since SARAH has limited computation due to it being a Mathematica package.

4.2 SPheno[3], [4]

The SARAH-generated SPheno module was compiled (SPheno 4.0.7, `gfortran`) and run on **BS1**, Table 4.1.

Parameter	Value
$\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	0.26, 0.5, 0.5, -0.5 , 8×10^{-11}
m_2^2	$1.85 \times 10^5 \text{ GeV}^2$
$M_N = \text{diag}(M_1, M_2, M_3)$	$\text{diag}(345, 4800, 6800) \text{ GeV}$
h_{ij}	full 3×3 complex matrix, $\mathcal{O}(0.01-1)$ (see [27])

Table 4.1: Input parameters of benchmark point **BS1** ($m_2^2 \rightarrow \text{mEt2}$, $h_{ij} \rightarrow \text{Yn}$, $M_N \rightarrow \text{Mn}$ in the SARAH/SPheno files).

Particle	Our result (GeV)	Ref. [27], BS1 (GeV)
h (SM Higgs)	125.548349	125.548349
η_R, η_I	430.116263	430.116263
$\eta^\pm$	447.388134	447.388134
N_1, N_2, N_3	345, 4800, 6800	345, 4800, 6800

Table 4.2: Tree-level mass spectrum at **BS1**, matching Ref. [27] to all quoted digits.

FLAVORKIT gives $\text{BR}(\mu \rightarrow e\gamma) = 1.02 \times 10^{-10}$, matching [27] and $\sim 10^3 \times$ above the MEG bound 4.2×10^{-13} , so **BS1** is experimentally excluded (used here only to validate the pipeline).

4.3 MicroOmegas[5]

micrOMEGAs 7.1.4 was built from the SARAH CalcHEP output and run on the BS1 spectrum. With N_1 (345 GeV) as the dark-matter candidate,

$$\Omega_{N_1} h^2 \simeq 0.925 \quad (x_f \simeq 24.3), \quad (4.1)$$

compared to $\Omega h^2 \simeq 1.08$ in [27] (difference might be due to micrOMEGAs version, 7.1.4 vs. 4.2.5), dominant channel $N_1 N_1 \rightarrow e_3 \bar{E}_3$ in both (27–28 %). Both values sit almost an order of magnitude above Planck, $\Omega_{\text{DM}} h^2 = 0.120 \pm 0.001$ [28], **BS1** thermally overabundant.

4.4 Madgraph[6]

The SARAH UFO¹ model was imported into MadGraph5_aMC@NLO 3.8.0 to generate

$$pp \rightarrow \eta_R \eta^+, \quad (4.2)$$

at tree level via s -channel W^+ , e.g. Fig. 4.1 (MadGraph’s own diagram output for the $u\bar{d} \rightarrow \eta_R \eta^+$ channel; the remaining three channels below are the same diagram with $u \rightarrow c$ and/or $d \rightarrow s$).

¹Universal FeynRules Output

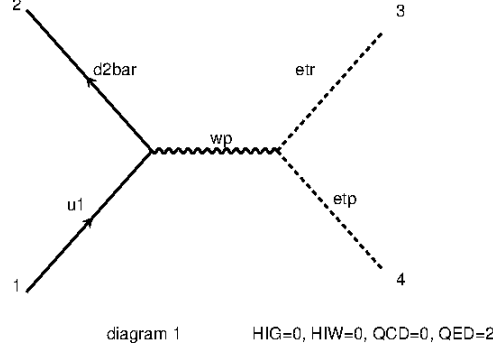


Figure 4.1: MadGraph5_aMC@NLO diagram for $u\bar{d} \rightarrow \eta_R \eta^+$ via s -channel W^+ .

(MadGraph’s auto-generated `param_card.dat` does not share SPheno’s block structure; directly overwriting it, as in [27], fails on this MadGraph version, so values were instead mapped block-by-block from the SPheno output.)

Channel	σ (fb)	Share
$u\bar{d} \rightarrow \eta_R \eta^+$	753.4	94.6 %
$u\bar{s} \rightarrow \eta_R \eta^+$	22.73	2.9 %
$c\bar{s} \rightarrow \eta_R \eta^+$	18.15	2.3 %
$c\bar{d} \rightarrow \eta_R \eta^+$	1.813	0.2 %
Total	796.1×10^{-3}	100 %

Table 4.3: Per-channel breakdown of $\sigma(pp \rightarrow \eta_R \eta^+)$ at BS1 (each with its charge-conjugate mirrored automatically by MadGraph); dominated by valence-quark PDFs, as expected.

Collider	$\sqrt{s}$	$\sigma(pp \rightarrow \eta_R \eta^+)$	Events generated
pp	13 TeV	0.7961 ± 0.0008 fb	10 000

Table 4.4: MadGraph run summary at BS1, against $\sigma = 0.7943 \pm 0.0008$ fb in [27].

At 300 fb^{-1} this gives $\mathcal{O}(240)$ events before acceptance/branching ratios.

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